

AIM: To design and simulate an IoT-based smart agriculture system using MATLAB by monitoring soil moisture levels, controlling water pump status (ON/OFF), and evaluating irrigation efficiency for optimized water resource management.

Objective:

To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing simulation in MATLAB.

Procedure:

1. Define soil moisture range (0–100%) and set an irrigation threshold value (e.g., 40%).
2. Generate or input soil moisture data to simulate field conditions.
3. Compare soil moisture with the threshold value to determine pump status (ON if low moisture, OFF if sufficient moisture).
4. Simulate water pumping action based on control logic.
5. Calculate irrigation efficiency based on moisture improvement and water usage.
6. Display soil moisture level, pump status, and efficiency results.
7. Plot graphs to analyze soil moisture variation and irrigation performance.
8. Evaluate system behavior for smart agriculture decision-making.

Matlab Code :

```
clc; clear; close all;
```

```
% Simulated soil moisture data (%)
```

```
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
```

```
% Threshold for irrigation
```

```
threshold = 40;
```

```
figure
```

```
% ----- Graph 1: Soil Moisture Variation -----
```

```

subplot(2,1,1)

plot(time, soil_moisture, '-o', 'LineWidth', 2)

title('Soil Moisture Monitoring in IoT Smart Agriculture')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

```

grid on

hold on

```

ylime(threshold,'r--','Threshold')

```

% ----- Graph 2: Moisture Status (Dry/Wet Level) -----

```

status = soil_moisture >= threshold;

```

```

subplot(2,1,2)

```

```

stem(time, status, 'filled', 'LineWidth', 2)

```

```

title('Soil Condition Status (1=Wet, 0=Dry)')

```

```

xlabel('Time (Hours)')

```

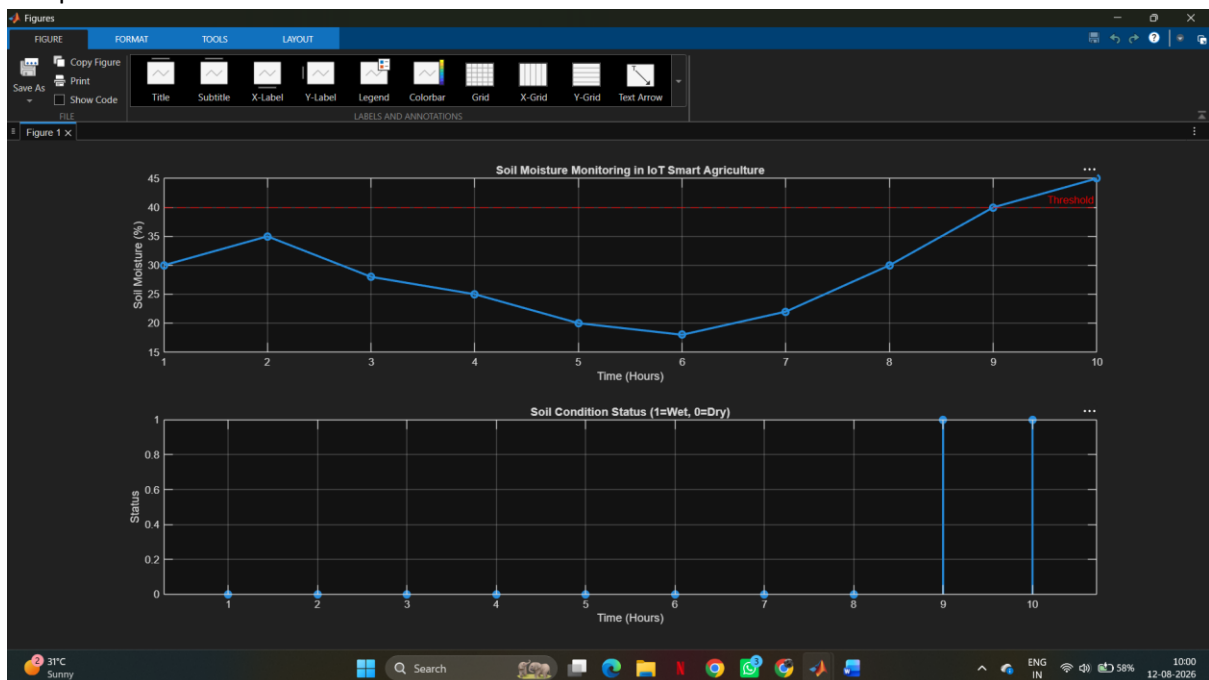
```

ylabel('Status')

```

grid on

Output:



Result:

1. The soil moisture levels were successfully monitored using MATLAB simulation, showing variation over time and identifying dry and wet conditions based on threshold values.
2. The water pump control system was effectively implemented, where the pump automatically switched ON when soil moisture dropped below the threshold and remained OFF when moisture was sufficient.
3. The irrigation efficiency was successfully calculated and analyzed, showing that the system optimizes water usage by maintaining soil moisture close to the desired level.